Optically written interface states at the pump boundary of a stretched-hexamer exciton-polariton lattice: a numerical study

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Abstract. We show numerically that a non-resonant pump spot, which acts only as a local blueshift of the exciton resonance, writes an interface into a lattice of stretched hexamers of etched micropillars in a single (In,Ga)As quantum-well microcavity, and that this interface carries localised polariton modes. Solving the four-component photon–exciton eigenproblem of a finite 252-pillar structure, we find for a hexagonal pump spot 18 polarisation doublets that lie between the pump-shifted interior band and the exciton line and carry 90–94 % of their photon intensity on the 18 pillars whose inter-cell partner lies outside the pump (Fig. 1). The interface modes are 69–76 % photonic and therefore bright in emission. A straight pump edge along a lattice direction produces the corresponding line-shaped interface state (Fig. 2). Varying the pillar radius, lattice constant and hexamer stretch one at a time, we determine the fabrication tolerance of the effect (Fig. 3). The pillar radius, the quantity most exposed to lithographic scatter, is the critical one: the ring survives from −2.5 % to at least +7.5 % of the design radius, degrades gradually for larger and abruptly for smaller pillars. Lattice constant and stretch are uncritical within ±5 % and ±60 nm.

1 Introduction

Lattices of hexamers whose six sites are pushed away from (“stretched”) or towards the cell centre are a photonic realisation of the Wu–Hu scheme for crystalline topological phases: whether the inter-cell or the intra-cell coupling dominates decides on which side of a band inversion the lattice sits, and boundaries between the two cases host localised states. In exciton-polariton systems the lattice can in addition be modified *in situ*. A non-resonant pump creates an exciton reservoir that blueshifts the exciton resonance locally; the polariton modes inside the pump spot shift by their exciton fraction, so the pump spot defines a region with shifted bands and a boundary that may host interface modes — without any change of the fabricated structure and with a geometry that can be reconfigured by shaping the pump beam.

Here we quantify this mechanism for a concrete design of a single-quantum-well sample. Section 2 summarises the model, the parameters and the figure of merit used to find the design. Section 3 shows the interface state written by a hexagonal pump spot (Fig. 1) and by a straight pump edge (Fig. 2), its tolerance against deviations of the lithographically defined geometry (Fig. 3, Fig. S1), and clarifies a sharp exciton feature on the pump boundary (Fig. S2). Section 4 lists the limitations of the model.

2 Model and numerical method

2.1 Hamiltonian

The cavity photon field is described in the linear basis (E_x, E_y), the quantum-well exciton in the circular basis (X_+, X_-). On a real-space grid the stationary problem reads

$$H = [\, -\hbar^2\nabla^2/2m_c + U_{ph}(\mathbf{r})\,]_{photon} \oplus [\, E_X - \hbar^2\nabla^2/2m_X + U_X(\mathbf{r})\,]_{exciton} + (\hbar\Omega_R/2)\,(T + T^\dagger),$$
$$\text{with the photon–exciton coupling } T = [[1, i], [1, -i]]/\sqrt{2} \text{ (linear} \rightarrow \text{circular basis),}$$

with TE–TM splitting, exciton Zeeman splitting and interface anisotropy set to zero. The etched lattice enters only through the photonic potential U_{ph}: the unetched mesas (pillars) sit at the bare cavity energy E_{free}, the etched barrier E_{conf} above it. The pump enters only through the exciton potential $U_X(\mathbf{r}) = B_{eff}$ inside the pump spot, with a Gaussian-softened edge (σ = 0.3 μm). B_{eff} is an effective parameter: it lumps the reservoir-induced exciton shift and the saturation-induced reduction of the Rabi splitting, which both raise the lower polariton, and is to be calibrated against a measured polariton shift at known Hopfield fraction.

2.2 Parameters

quantity	value	origin
exciton energy E_X	1.4707 eV	QW calibration sample, one $In_{0.06}Ga_{0.94}As$ QW
Rabi splitting $\hbar\Omega_R$	2.45 meV	3-QW sample (4.16 meV) scaled by $1/\sqrt{3}$, slightly increased
photon mass m_c	$3.2\times10^{-5}\, m_e$	fit of a pillar-overlap series, transfer-matrix consistent
exciton mass m_X	0.20 m_e	standard value
photon linewidth	0.20 meV FWHM	Lorentzian (cavity loss)
exciton linewidth	1.5 meV FWHM	Gaussian, inhomogeneous; QW calibration sample
confinement depth E_{conf}	19.92 meV	design (search range 10–20 meV)
mesa detuning $E_{free} - E_X$	−4.62 meV	design ($-1.886\ \hbar\Omega_R$)
pillar radius r	0.958 μm	design
lattice constant a_{latt}	6.773 μm	design ($a_{latt} = 6rv$, overlap v = 1.178)
stretch s	111.2 nm	design; intra-cell centre distance 2.369 μm, inter-cell 2.035 μm
effective pump blueshift B_{eff}	2.61 meV	design (search range 0–3 meV)
hexagonal pump spot	apothem 8.87 μm	1.31 a_{latt}, centred on a hexamer, edge blur σ = 0.3 μm

With these numbers neighbouring pillars of adjacent cells are separated by a gap of 0.12 μm, pillars within a cell by 0.45 μm: the inter-cell coupling dominates, i.e. the hexamers are in the expanded configuration. B_{eff} = 2.61 meV exceeds the 2.30 meV estimated for the exciton shift alone; effective shifts of up to about 6 meV are considered reachable (Sec. 4).

2.3 Numerics and figure of merit

The operator is discretised with second-order finite differences on a 147 nm grid (the 150 nm target adjusted to the lattice period); the potentials are supersampled tenfold before averaging onto the grid. For the finite window of $N = 252$ pillars we compute the lowest $2N$ eigenmodes (the *s* band, two polarisations per pillar) with a sparse GPU eigensolver. Real-space spectra ("tomographies") are built from the photon components, as observed in emission, with a Voigt line per mode: Lorentzian half width $|C|^2$ times the photon half width, Gaussian full width $|X|^2$ times the exciton full width. The edge pillars are defined geometrically as the pillars inside the pump spot whose inter-cell (nearest-neighbour) bond is cut by the pump boundary. The figure of merit is

$$cost = min_E \langle I_{ph} \rangle_{bulk}(E) / \langle I_{ph} \rangle_{edge}(E),$$

where $\langle \cdot \rangle$ is the mean photon intensity in 0.5 µm Gaussian windows centred on the edge or on all other pillars, evaluated on every second point of a 100-point energy axis. A small cost means that an energy E^* exists at which the edge pillars outshine the rest of the lattice. In the search, energies were admitted only if the mean edge intensity exceeded a fixed absolute value (5×10^{19} in the units of the tomography). For the tolerance analysis we additionally use a *threshold-free* cost, the same ratio minimised over the energies at which the edge intensity reaches at least 10 % of its own maximum; it removes the dependence on the absolute intensity scale (Sec. 3.3). The design was found by a global search (dual annealing, 2247 evaluations) refined by a local Nelder–Mead search (62 evaluations), both on a 350 nm grid for speed. On the 147 nm grid the same design has a cost of 0.091 instead of 0.056: the mode structure is unchanged, the contrast is grid sensitive, and all results below are given on the finer grid.

3 Results

3.1 A ring-shaped interface state written by a hexagonal pump spot

The hexagonal pump spot covers 42 of the 252 pillars and cuts 18 inter-cell bonds; the inner partners of these bonds are the edge pillars (Fig. 1a). The spectrum of the 504 lowest modes (Fig. 1b) splits into four groups. The unpumped lattice forms the *s* band 1.48–1.00 meV below E_X with exciton fractions 0.41–0.60. Inside the pump spot the exciton is shifted up by B_{eff}; the corresponding modes are strongly photonic ($|X|^2 \approx 0.11$) and lie 0.89–0.74 meV below E_X. Above them, 0.61–0.41 meV below E_X, follow 36 modes (18 polarisation doublets) with $|X|^2 = 0.24$–0.31 that carry 90–94 % of their photon intensity on the 18 edge pillars, which make up only 7 % of all pillars. These are the interface modes. A dense cluster of exciton-like states ($|X|^2$ up to 0.96) just below E_X carries no weight on the edge.

The interface modes appear as a pronounced peak of the mean edge intensity, where the bulk intensity has already dropped (Fig. 1c); the bulk-to-edge ratio reaches its minimum of 0.091 at $E^* - E_X = -0.53$ meV. Constant-energy cuts of the photon channel through the interface band (Fig. 1d–f) show the closed ring on the 18 edge pillars at all three energies, with a weak contribution of the pumped interior that fades towards the top of the band. Because the interface modes are 69–76 % photonic and lie inside the gap between the interior band and the exciton-like cluster, the ring is directly observable in real-space photoluminescence at energies 0.4–0.6 meV below E_X. In a tight-binding reading, the edge pillars are the members of the strongly coupled inter-cell dimers whose partner lies outside the pump, at an exciton energy lower by B_{eff}; deprived of the bonding with their partner they are pushed above the interior band.

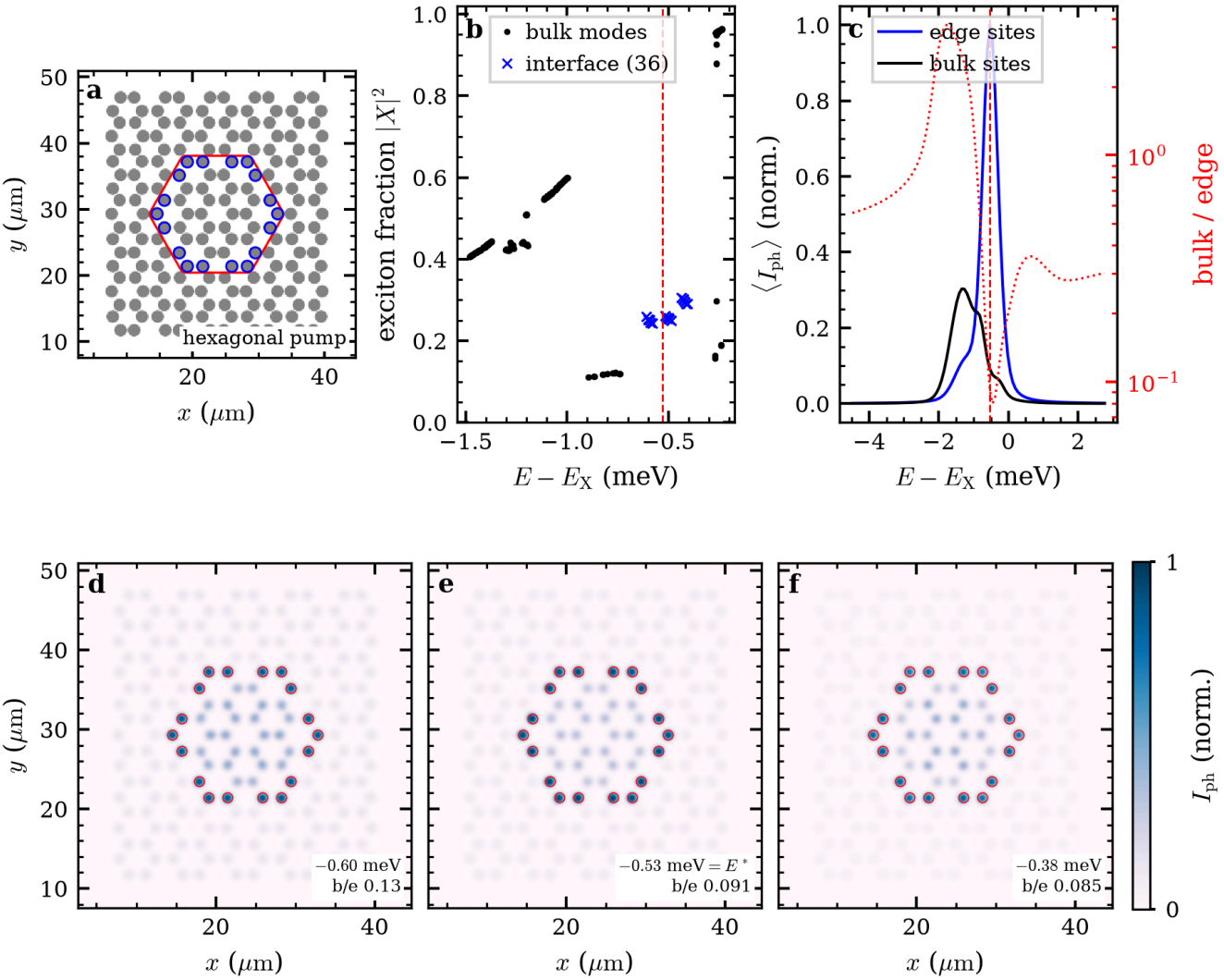


Figure 1. A hexagonal pump spot writes a ring-shaped interface state. a, Lattice layout (grey: pillars of the photonic potential), pump-spot boundary (red, half maximum of U_X) and the 18 edge pillars (blue circles), i.e. the pillars inside the pump whose inter-cell partner is outside. **b**, Exciton fraction of the 504 lowest eigenmodes versus energy; blue crosses: interface modes (more than half of the photon intensity on the edge pillars), black dots: all other modes; red dashed: energy E^* of minimal cost. **c**, Mean photon intensity on the edge pillars (blue) and on all other pillars (black) versus energy, and their ratio (red dotted, right axis), the quantities that define the cost. **d–f**, Photon-channel tomography at the lowest interface-mode energy (d), at E^* (e) and at the highest interface-mode energy (f), on a common intensity scale; labels give $E - E_X$ and the bulk-to-edge ratio at that energy (the cost is evaluated on every second energy point, hence f can fall slightly below e). Parameters of Sec. 2.2, 147 nm grid.

3.2 Straight pump edge

The interface does not rely on the closed hexagonal shape. In Fig. 2 the pump covers the half-plane to one side of a straight edge running at 60° to the x axis, i.e. parallel to a lattice vector, placed on a boundary between hexamer columns so that only whole unit cells are pumped (Fig. 2a). The edge cuts two inter-cell bonds per cell row, leaving 13 edge pillars along the line. The spectrum (Fig. 2b) has the same structure as for the hexagon: 26 interface modes 0.57–0.43 meV below E_X with $|X|^2$ = 0.24–0.28 and 87–93 % of their photon intensity on the edge pillars. The photon cuts (Fig. 2d–f) show the line-shaped interface state along the full edge. The cost, 0.176 at $E^* - E_X$ = −0.51 meV, is twice that of the hexagon. The main reason is visible in the cuts: where the pumped half-plane reaches the outer boundary of the simulated lattice, the termination of the lattice acts as a second interface, and the outermost pumped pillars (right-hand column)

light up at the same energy. These pillars count as bulk in the cost. In an experiment this is avoided by keeping the pump spot inside the lattice.

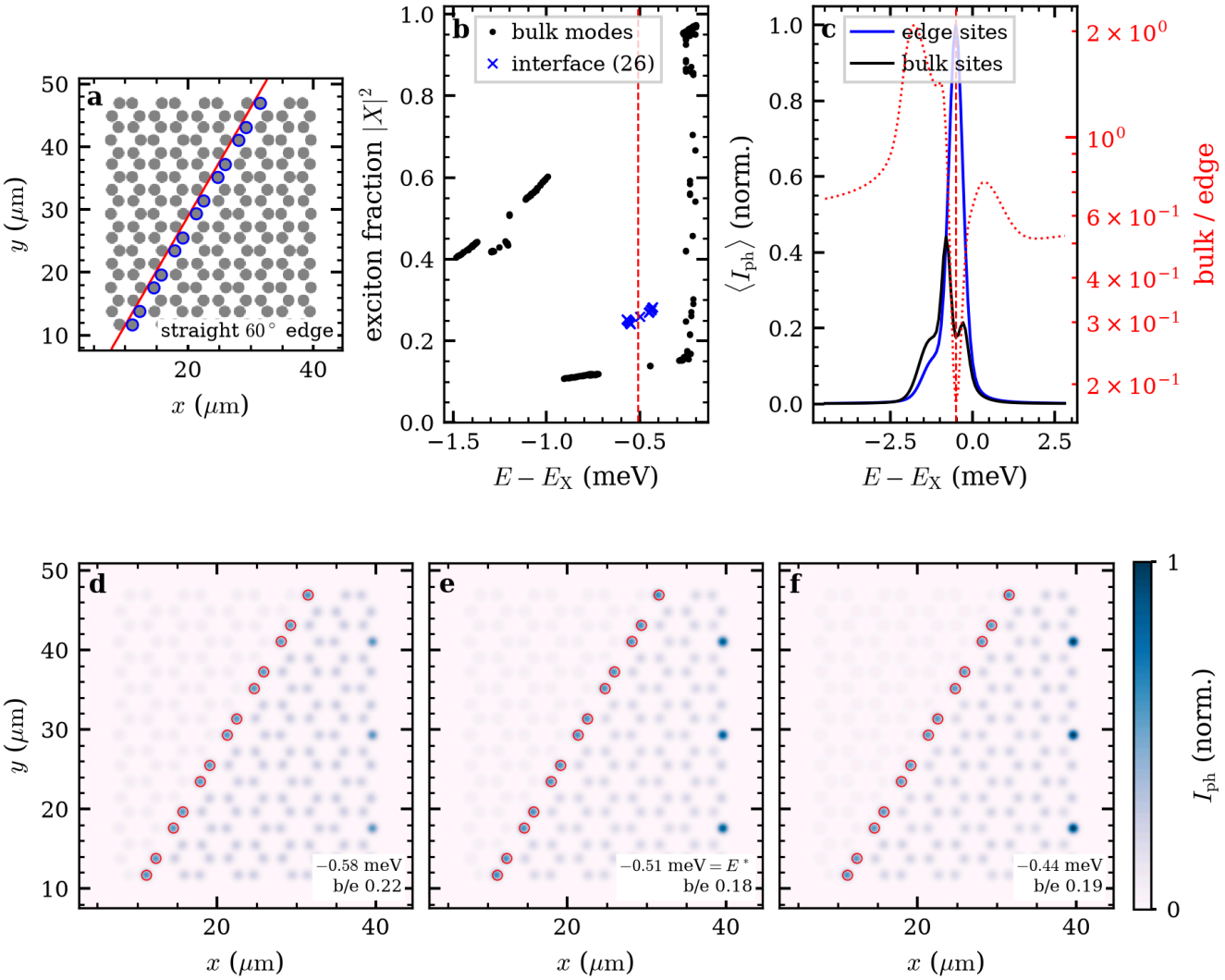


Figure 2. A straight pump edge along a lattice direction carries a line-shaped interface state. Panels as in Fig. 1 for a pump covering the half-plane to the lower right of a straight edge at 60° (red line in a), placed on the hexamer-column boundary nearest to the drawn target line; 147 of the 252 pillars are pumped, 13 are edge pillars. **d–f**, Photon-channel cuts at the lowest interface-mode energy, at E^* and at the highest interface-mode energy. Besides the edge line, the outermost pumped pillars at the right boundary of the simulated lattice are bright: the lattice termination inside the pumped region acts as a second interface. All other parameters as in Fig. 1.

3.3 Fabrication tolerance

The pillar radius is the parameter most likely to deviate in the fabricated sample, because it depends on the electron-beam exposure dose and on development and etching; the mesa positions, which set the lattice constant and the stretch, are defined by the beam deflection and are considerably more stable. We therefore vary the radius at fixed pillar centres (the overlap $v = a_{\mathrm{latt}}/6r$ follows) over the widest range, ±10 %, and the lattice constant (±5 %, radius fixed) and the stretch (±60 nm) for completeness, each alone and with the hexagonal pump of Fig. 1.

The cost used in the search turns out to be unsuited for this purpose without modification. At the design point the mean edge intensity peaks only 10 % above the admission threshold of the search (5.5 versus 5.0×10^{19}): the optimiser has converged to the corner where the edge signal barely passes the threshold. Already a 2.5 % smaller radius lowers the peak below it on the evaluated energy points, and the cost

formally becomes infinite although the interface ring is intact and its contrast even improves. We therefore judge the tolerance with the threshold-free cost of Sec. 2.3, which at the design point is 0.085 (the thresholded value is 0.091), and call the interface mode usable as long as this cost stays below twice its design value. Figure 3 shows both costs, the energy of minimal cost and the photon fraction of the interface modes for all scan points; Fig. S1 shows the photon maps at that energy for every point, so that "usable" can also be judged by eye.

Pillar radius. The cost stays within the bound from 0.910 to 1.030 μm, i.e. from −5.0 % to +7.5 % (−48 to +72 nm) around the design radius 0.958 μm (Fig. 3a). The two sides behave differently. Larger pillars at fixed positions increase the overlap of neighbouring pillars; the interface band moves down, to 0.93 meV below E_X at +10 % (Fig. 3d), its photon fraction stays at 0.71–0.73 (Fig. 3g), and the contrast degrades gradually because the pumped interior lights up together with the ring (Fig. S1, top row). Smaller pillars raise all photonic energies towards the exciton. At −2.5 % the ring is intact and even cleaner than at the design (cost 0.054), 0.16 meV below E_X. At −5 % the cost is still within the bound, but only 23 interface modes remain, they are 60 % excitonic and lie at the exciton line, and the ring visibly fades; below −7.5 % no interface modes are left. A practically usable radius window is therefore about −2.5 % to +7.5 % (0.934–1.030 μm): undersized pillars are critical, oversized ones are tolerated. For fabrication this suggests aiming at the upper half of the window.

Lattice constant and stretch. Both are uncritical within the scanned ranges. Over ±5 % of the lattice constant (6.43–7.11 μm) the cost stays between 0.077 and 0.100, all 36 interface modes persist, and their photon fraction rises from 0.62 to 0.79 with increasing lattice constant (Fig. 3b, e, h). Over ±60 nm of stretch (51–171 nm) the cost stays between 0.065 and 0.105; larger stretch gives a cleaner ring (lowest cost 0.065 at 141 nm), at the price of a lower photon fraction, which falls from 0.80 to 0.61 (Fig. 3c, f, j; Fig. S1, bottom row). Neither parameter reaches the bound inside the scan, so the true tolerances are wider than ±5 % and ±60 nm. The thresholded cost of the search becomes infinite for 13 of the 25 points, including several with an intact, high-contrast ring (Fig. 3a–c), which confirms that it should not be used to judge tolerances.

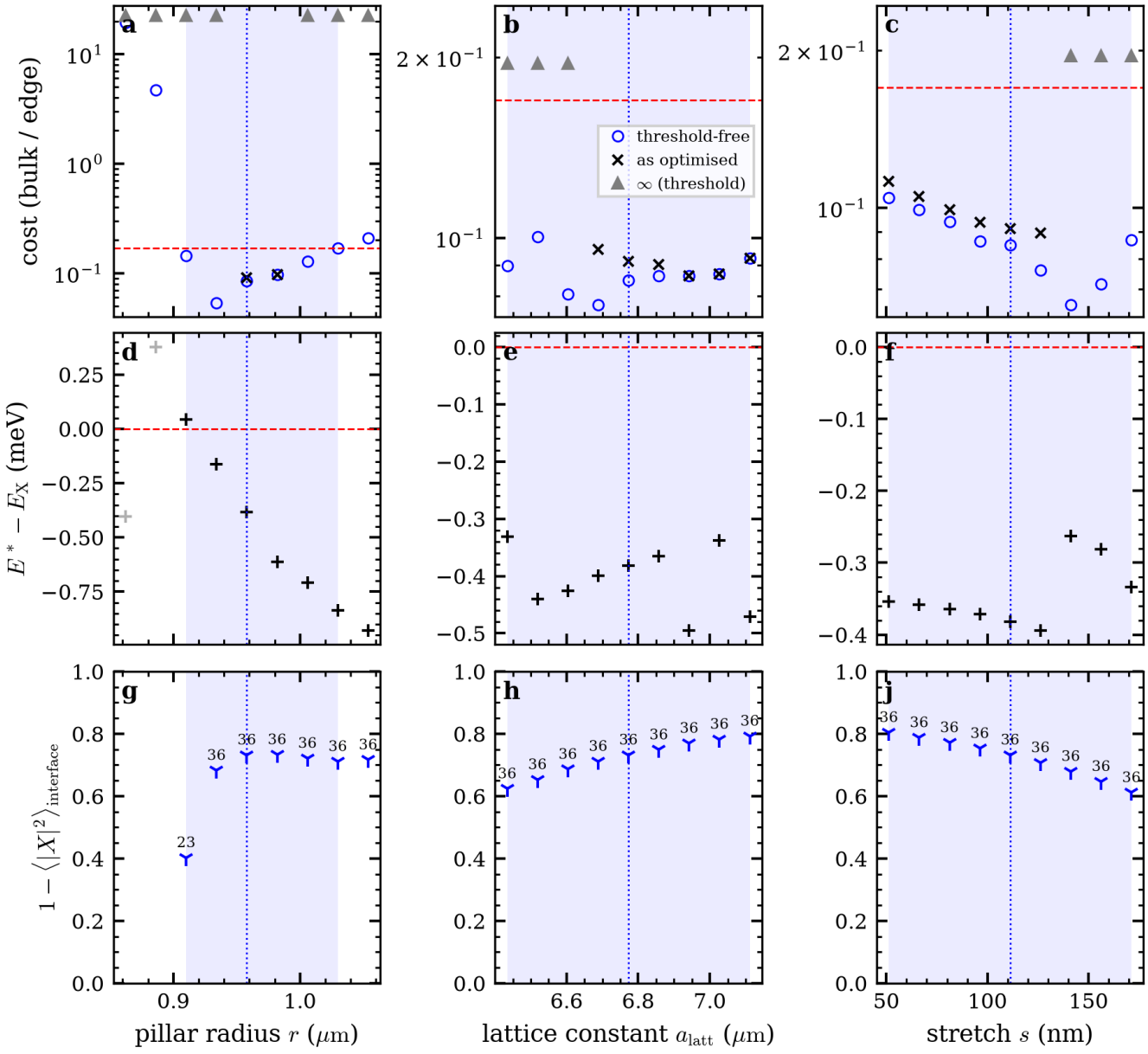


Figure 3. Tolerance of the interface mode against the three lithographically fixed parameters. a–c, Cost versus pillar radius (lattice constant fixed), lattice constant (radius fixed) and stretch, each varied alone around the design (blue dotted line): threshold-free cost (blue circles), cost as used in the search (black crosses; grey triangles mark points where it is infinite because the edge intensity stays below the admission threshold). Red dashed: twice the threshold-free design cost; shaded: contiguous range within this bound. **d–f**, Energy of minimal threshold-free cost relative to E_X (grey: no interface modes left). **g, h, j**, Mean photon fraction of the interface modes; numbers give how many modes carry more than half of their photon intensity on the edge pillars. Hexagonal pump, 147 nm grid, all other parameters as in Fig. 1.

3.4 Exciton states on the pump edge

In the exciton channel, energies between E_X and $E_X + B_{\text{eff}}$ show a sharp line that traces the pump boundary itself rather than the pillars, for example at $E - E_X = 1.08$ meV for the hexagonal spot, while the photon channel shows nothing there. These states are unrelated to the lattice: the exciton feels only the pump potential, which rises from 0 to B_{eff} across the softened edge. Because the exciton is heavy, its kinetic energy on the grid is tiny, and an exciton of energy E can only exist where $U_X(\mathbf{r}) \approx E - E_X$, on a closed isoline of the pump profile.

Figure S2 confirms this. For the modes of the exciton-only operator on the hexagonal pump map, the mean pump potential equals the mode energy to within 49 μeV over the whole window (Fig. S2b); the modes are a single grid cell wide and move from 0.49 μm outside to 0.46 μm inside the nominal pump line as the energy increases (Fig. S2a). A one-dimensional cut through the edge with successively finer grids identifies the origin. The discretised exciton band has the width $4\hbar^2/(2m_X\Delta x^2)$, only 34 μeV at 150 nm. As long as this is much smaller than B_{eff}, the states of a given energy are confined to the iso-line (Fig. S2c). Once the grid resolves the exciton dispersion over the pump window ($\Delta x \leq 19$ nm), the same energies host freely propagating, photon-dark excitons outside the pump, and only an enhancement of the local density of states remains at the classical turning point (Fig. S2d), with the Airy length $(\hbar^2/2m_X|\nabla U_X|)^{1/3} = 38$ nm at the steepest part of the edge. The bright exciton line therefore marks the turning-point region of the dark exciton continuum, sharpened into a one-cell-wide line by the grid. It is neither a lattice nor a topological edge state and does not affect the polariton interface modes.

4 Discussion and limitations

- **Grid convergence.** Between the 350 nm search grid and the 147 nm grid the cost changes from 0.056 to 0.091 and E^* by 0.3 meV, while the mode structure and the ring are unchanged. Quantitative statements refer to 147 nm; a further refinement step has not been done.
- **Effective pump.** The pump is a static exciton shift B_{eff}; reservoir dynamics, gain and polariton–polariton interaction are not included. $B_{\mathrm{eff}} = 2.61$ meV exceeds the 2.30 meV estimated for the exciton shift alone. Since B_{eff} also contains the saturation-induced collapse of the Rabi splitting, considerably larger effective shifts, of the order of 6 meV, may be reachable in single (In,Ga)As quantum-well microcavities (estimate of the experimental group); the design value is not expected to be limiting.
- **Spectral window.** The cost and Figs. 1–3 use the lowest $2N$ modes. States above E_X require a separate shift-invert solve; they are not needed for the interface modes, which lie below E_X.
- **Exciton discretisation.** On the 147 nm grid the discrete exciton band is much narrower than B_{eff}, which produces the one-cell-wide exciton lines on the pump edge (Sec. 3.4). They are photon-dark and do not enter the cost.
- **Linewidths.** Photon and exciton broadening enter only the spectra, not the eigenmodes. The interface band (0.2 meV wide) is separated from the interior band by 0.13 meV and from the exciton-like cluster by 0.14 meV, comparable to the polariton line widths; the contrast in Figs. 1c and 2c already includes this broadening.
- **Topological character.** The interface modes are identified by their spatial localisation on the broken-dimer pillars; no topological invariant is computed here.
- **Finite structure.** A pump region that touches the boundary of the lattice creates additional boundary states there (Fig. 2); pump spots should stay inside the lattice.

5 Conclusion

A pump spot acting only on the exciton writes an interface into a lattice of expanded hexamers, and the broken-dimer pillars along its boundary carry a band of bright, 69–76 % photonic interface modes 0.4–0.6 meV below the exciton line, between the pump-shifted interior band and the exciton-like states. The effect follows the shape of the pump: a hexagonal spot gives a closed ring on its 18 edge pillars, a straight edge along a lattice direction a line. The design tolerates the expected lithographic deviations: the pillar radius may deviate by −2.5 % to +7.5 % (about −24 to +72 nm), with undersized pillars being the critical

direction, while the more stable mesa positions — lattice constant and stretch — are uncritical over ±5 % and ±60 nm. Two caveats qualify these statements: the contrast is not yet grid converged, and the effective pump shift B_{eff} = 2.6 meV remains to be calibrated on the sample. For an experiment we recommend a slightly oversized pillar radius and a pump spot that stays inside the lattice.

Supplementary figures

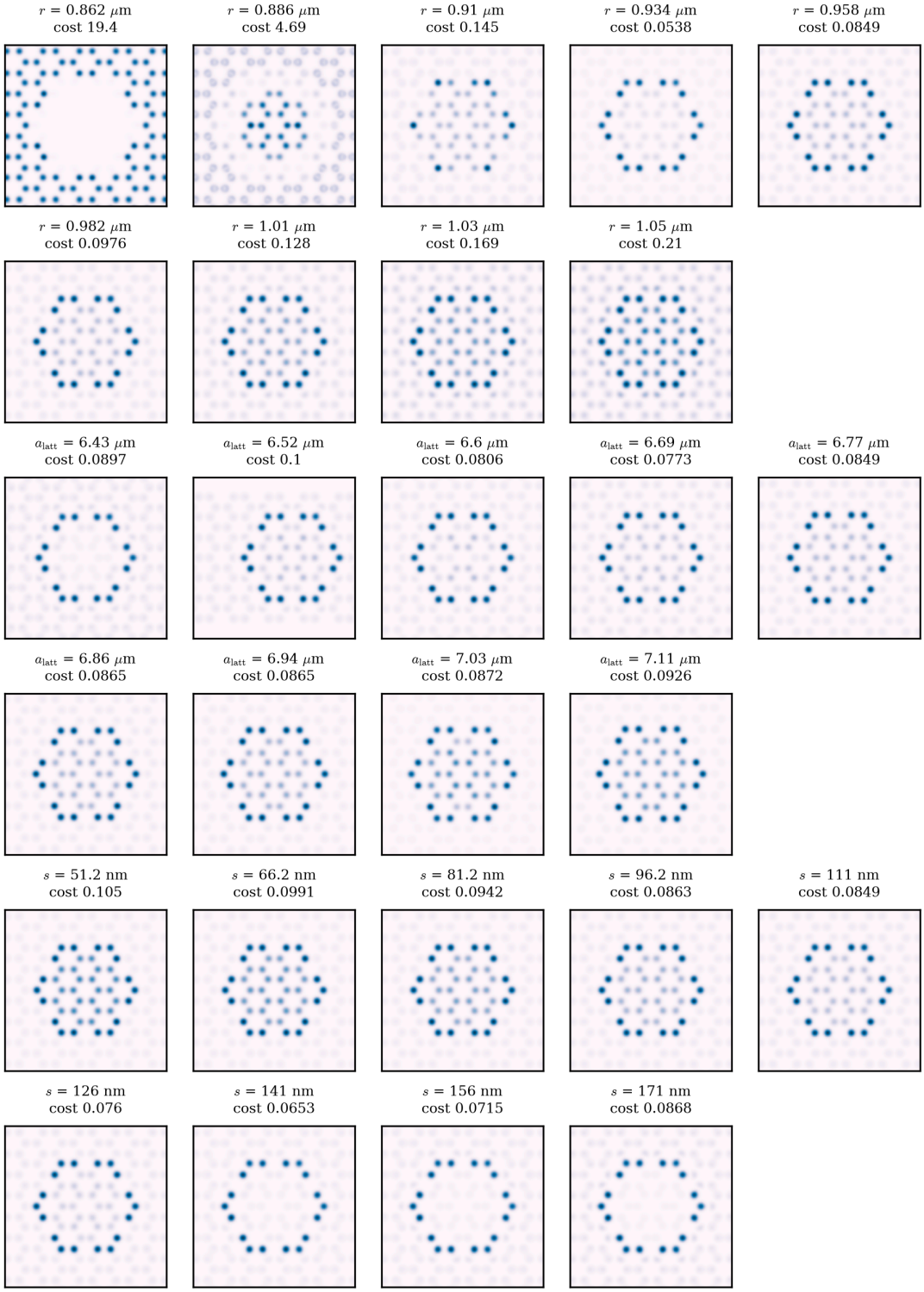


Figure S1. Photon tomography at E^* for every point of the tolerance scan of Fig. 3. Rows 1–2: pillar radius, rows 3–4: lattice constant, rows 5–6: stretch, each ordered by parameter value; titles give the parameter value and the threshold-free cost. Maps are taken at E^* of the thresholded cost, or, where that cost is infinite, at the energy of minimal threshold-free cost; each map is normalised to its maximum.

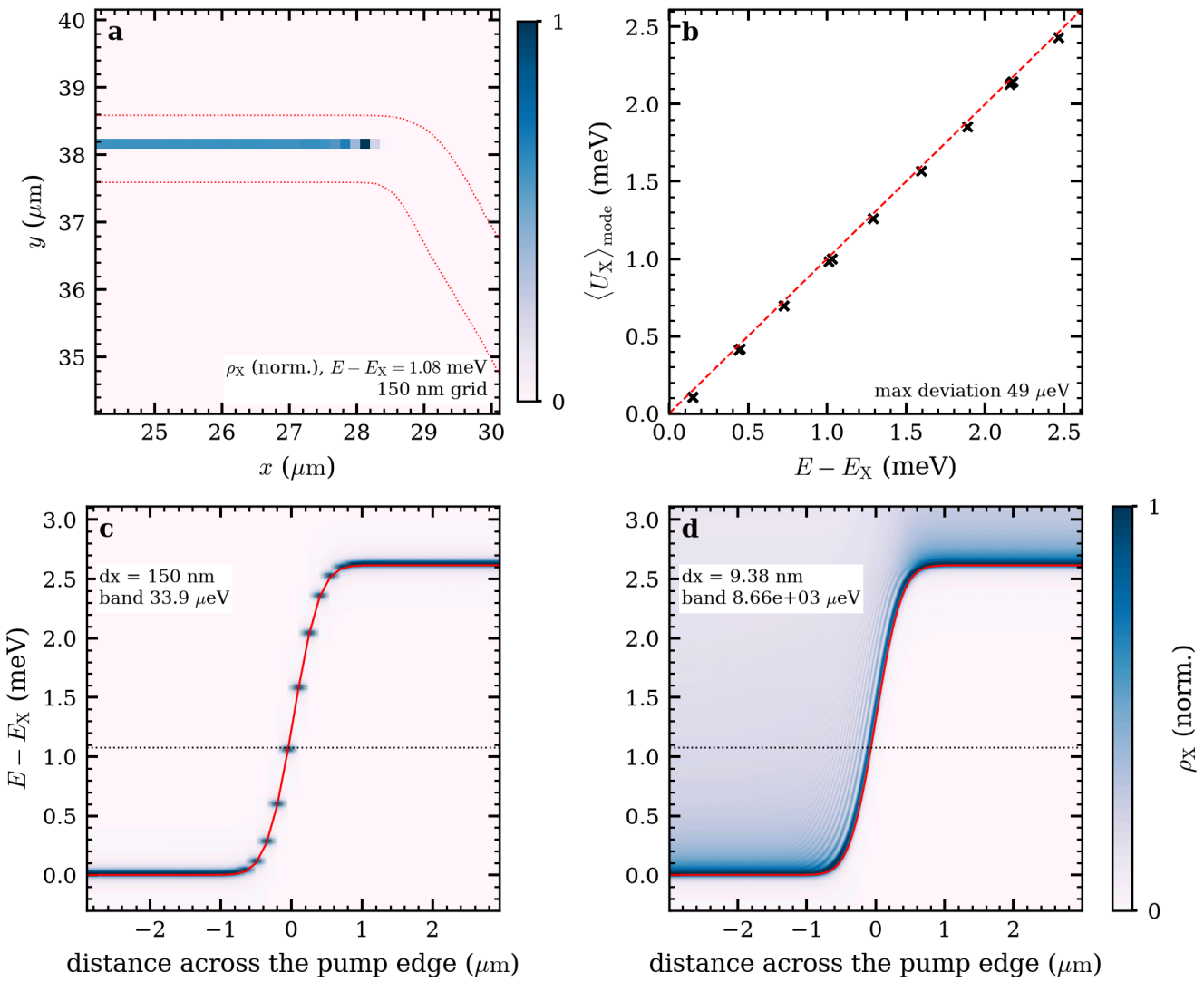


Figure S2. The exciton line on the pump boundary is a turning-point effect of the dark exciton continuum, sharpened by the grid. a, Local exciton density of states of the exciton-only operator on the hexagonal pump map at $E - E_X$ = 1.08 meV (60 modes nearest this energy, 150 nm grid), zoomed on one corner of the hexagon; dotted: 5 % and 95 % levels of U_X, i.e. the extent of the softened pump edge. **b**, Mean pump potential of each mode versus its energy (90 modes at nine energies); dashed: $\langle U_X \rangle = E - E_X$. **c, d**, Local exciton density of states across a straight pump edge with the same profile (one-dimensional cut) on a 150 nm (c) and a 9.4 nm (d) grid; red: $U_X(x)$; dotted: 1.08 meV. Exciton-only calculations; the photon channel shows no counterpart.

Reproduction (folder numerical_report): simulate_report_data.py (data of Figs. 1, 2), robustness_scan.py (Figs. 3, S1), exciton_contour_check.py (Fig. S2), plot_report_figures.py (all figures). Drivers gpe_v4.py (hexagonal pump) and gpe_v5.py (straight edge) in the parent folder; parameter search gpe_v3_optimise_fixed.py and gpe_v3_minimize.py.